

UNIVERSITY OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE

University Management System (UMS)

Project Proposal & Documentation

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1 Introduction

1.1 Background

Universities today face significant challenges in managing student finances, particularly when dealing with different student categories. Traditional manual methods using spreadsheets or paper records lead to calculation errors, delayed fee vouchers, and lack of category-wise financial reporting.

1.2 Purpose of the System

The University Management System aims to provide a digital solution that:

- Automatically calculates expenditures for hostel and day scholar students
- Generates accurate fee vouchers based on student category
- Provides PDF reports for both categories independently
- Reduces manual data entry errors by 90%
- Offers real-time financial summaries to administrators

2 Problem Statement

Universities currently manage hostel and day scholar expenditures manually using spreadsheets or paper records. This leads to:

- Calculation errors in fee vouchers
- Delayed fee generation process
- Lack of category-wise financial reporting
- Staff spending excessive time identifying applicable fee heads
- No automated distinction between hostel and day scholar expenses
- Students receiving incorrect vouchers with wrong charges
- Administrators unable to quickly generate category-wise revenue summaries

Therefore, a digital solution is urgently needed to address these challenges.

3 Objectives

The primary objectives of this project are:

1. Develop a University Management System that automatically calculates expenditure separately for hostel and day scholar students
2. Generate accurate fee vouchers based on student category
3. Provide PDF reports for both categories independently

4. Reduce manual data entry errors by 90%
5. Provide real-time financial summaries to university administrators
6. Ensure all expenditure heads are properly stored and easily updatable

4 Scope

The scope of this project includes:

- Student registration with category selection (hostel or day scholar)
- Hostel management for allocating rooms to hostel students
- Separate expenditure heads for hostel students (rent, mess, utilities, maintenance, security deposit)
- Separate expenditure heads for day scholars (transport, library, sports fees)
- Automatic fee voucher generation based on student category
- PDF reports for individual students and category-wise summaries

The system does **not** handle payroll, employee management, or university examination grading beyond fee calculation.

5 System Design

5.1 Merit Calculation Formula

Total Merit Percentage is calculated using the following weighted formula:

$$TotalMerit = \left(\frac{Matric}{1100} \times 25 \right) + \left(\frac{Inter}{1100} \times 45 \right) + \left(\frac{EntryTest}{100} \times 30 \right) \quad (1)$$

This formula ensures that:

- Intermediate performance carries the highest weight (45%)
- Entry Test contributes 30%
- Matriculation contributes 25%

5.2 Database Tables

Table Name	Description
Programs	ProgramID, ProgramName, DurationYears, TotalSeats, ClosingMerit
Students	StudentID, Name, RollNo, CNIC, ProgramID, Student-Category, Marks
Hostels	HostelID, HostelName, Address, TotalRooms, Warden-Name

Rooms	RoomID, HostelID, RoomNumber, Capacity, Rent, IsAvailable
HostelExpenditureHeads	HeadID, HeadName, Amount, IsRefundable
DayScholarExpenditureHeads	HeadID, HeadName, Amount, IsOptional
CommonExpenditureHeads	HeadID, HeadName, Amount, ApplicableToAll
StudentMerit	MeritID, StudentID, Score, Rank, GenerationDate
FeeVouchers	VoucherID, StudentID, TotalAmount, GenerationDate, DueDate
FeeVoucherDetails	DetailID, VoucherID, HeadID, HeadType, Amount
HostelAllocations	AllocationID, StudentID, RoomID, AllotmentDate, IsActive
Payments	PaymentID, VoucherID, AmountPaid, PaymentDate, PaymentMethod

Table 1: Database Tables Summary

6 Functional Modules

6.1 Admin Dashboard (6 Core Modules)

1. **Student Management (CRUD):** Add, update, delete, and view student profiles with category selection
2. **Program & Merit Management:** Manage academic programs and generate merit lists
3. **Hostel Setup:** Manage hostels, rooms, capacity, and room availability
4. **Fee Heads Configuration:** Define costs for hostel, day scholar, and common fees
5. **Fee Voucher System:** Generate vouchers and record payments
6. **Reports Center:** Generate category-wise reports in PDF format

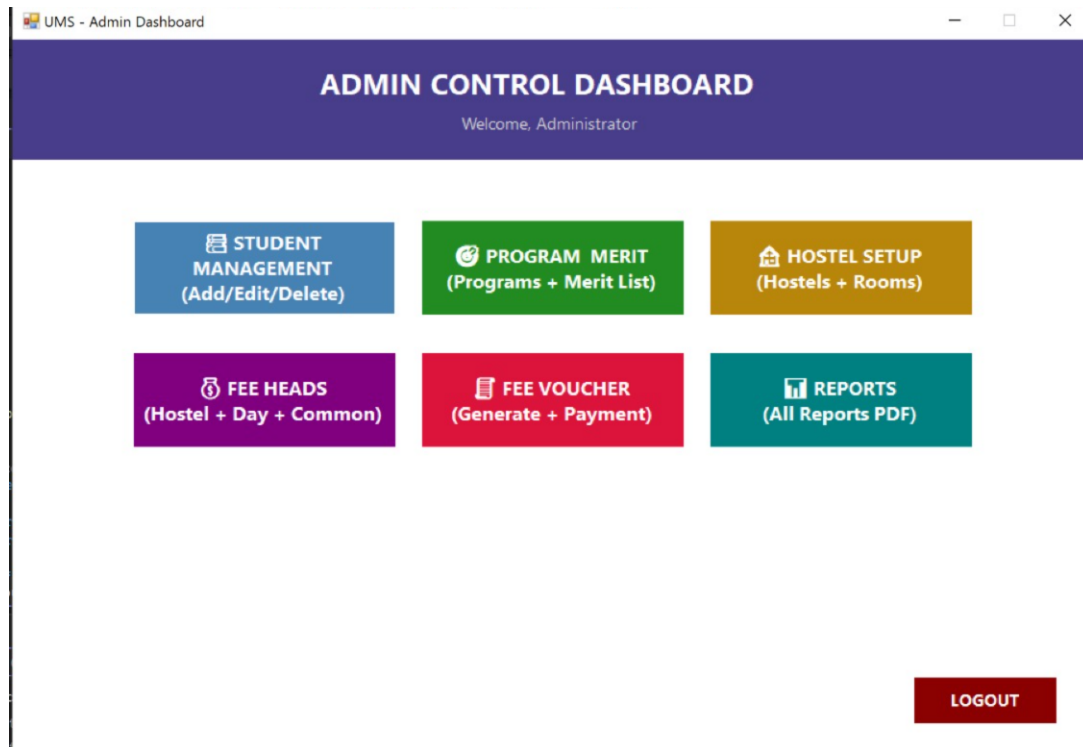


Figure 1: Admin Dashboard Interface

6.2 Accounts Dashboard

- View Vouchers: View and print fee vouchers
- Pending Fees: Generate pending fee and refund reports

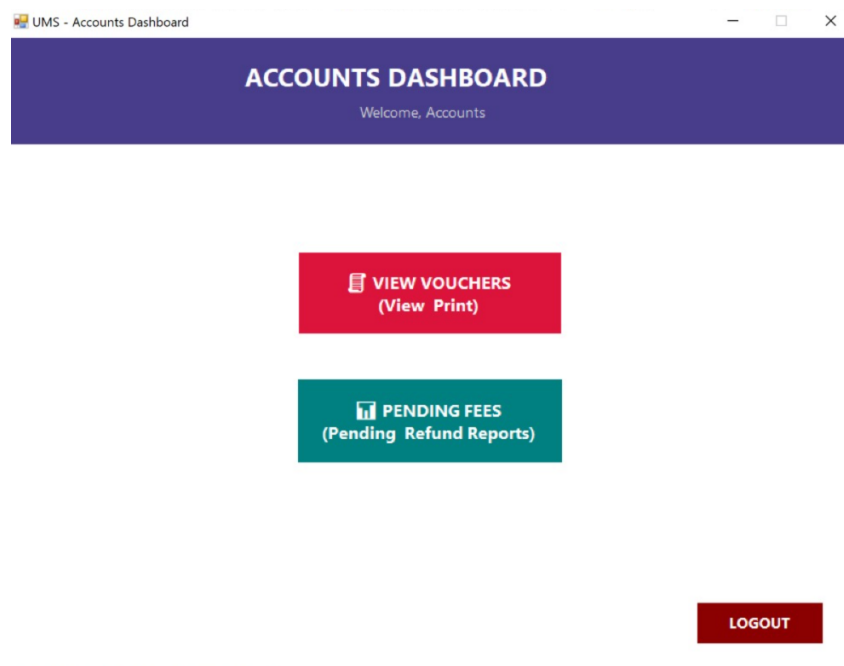


Figure 2: Accounts Dashboard Interface

6.3 Student Portal

- My Fee Voucher: Download current fee voucher as PDF
- Payment History: View past payment records
- My Allocation: View hostel room details (if applicable)

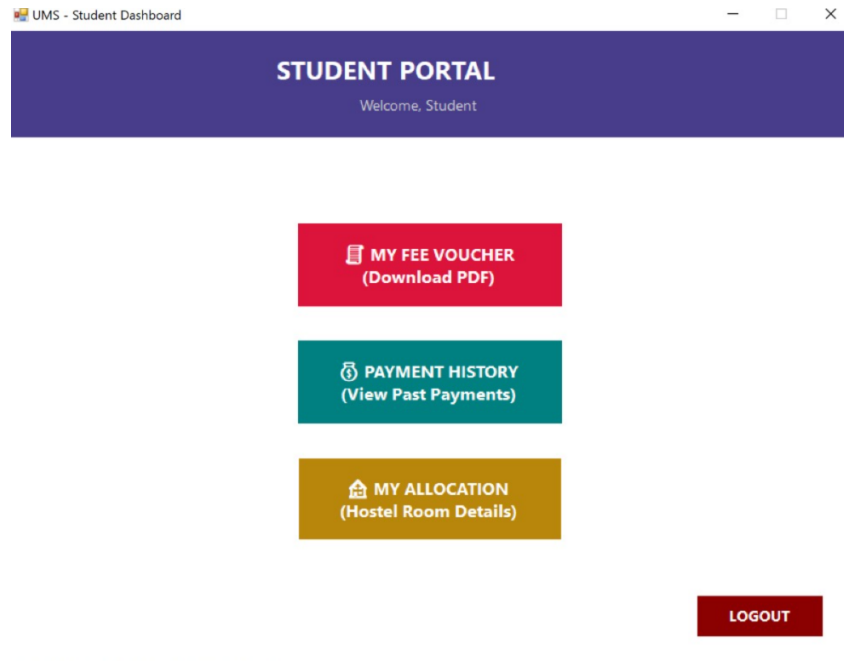


Figure 3: Student Portal Interface

7 Module Details

7.1 Student Payment History

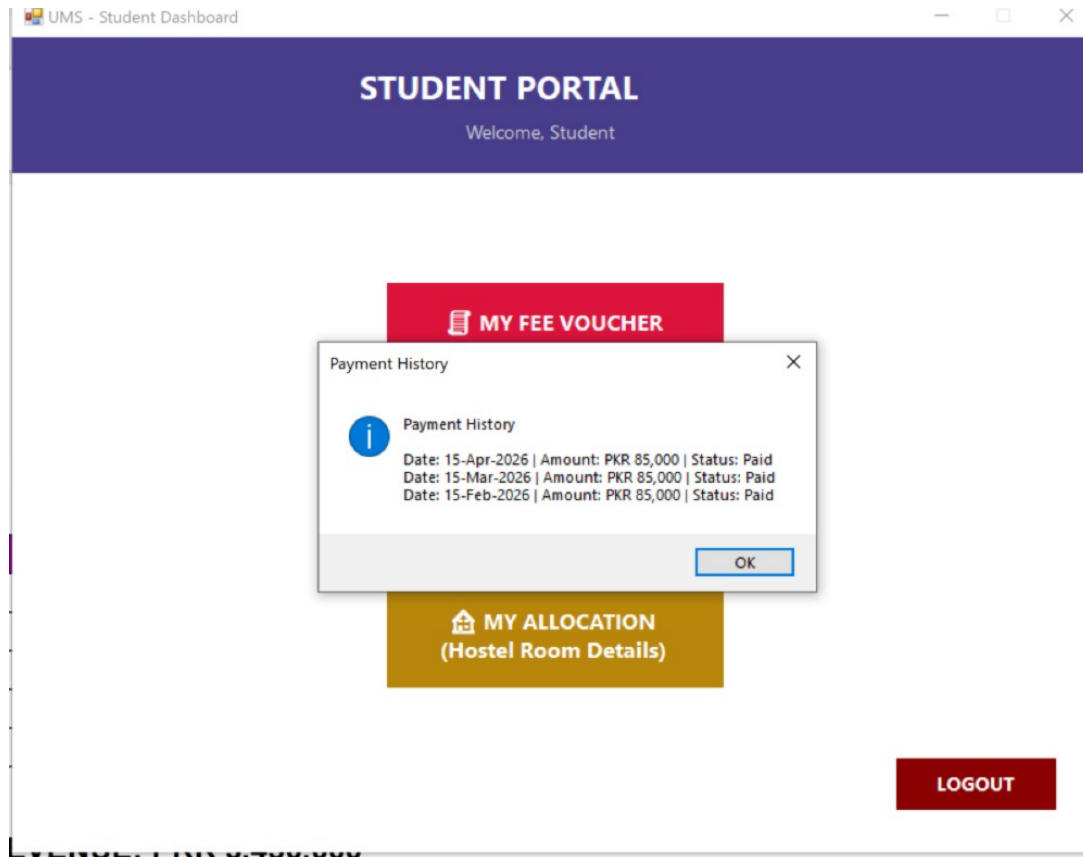


Figure 4: Student Payment History Form

Features:

- Add new students with complete profile
- Select category (Hostel/Day Scholar)
- Assign room number directly for hostel students
- Enter Matric, Inter, and Entry Test marks
- Search and edit existing student records

7.2 Hostel Setup Module

UMS - Hostel Setup

HOSTEL SETUP

Manage Hostels | Manage Rooms

Hostel: ADD ROOM

Room:

Capacity: VIEW OCCUPANCY

Rent:

Hostel	Room No	Capacity	Rent (PKR)
Girls Hostel A	229	2	8000
Girls Hostel A	227	8	4000

Figure 5: Room Management Tab

7.3 Fee Heads Configuration

UMS - Fee Heads Configuration

FEE HEADS

HOSTEL FEE HEADS | DAY SCHOLAR FEE HEADS | COMMON FEE HEADS

Hostel Expenditure Heads

Room Rent:

Mess Charges:

Utilities:

Maintenance:

Security Deposit:

SAVE HOSTEL

Figure 6: Hostel Fee Heads Configuration

The screenshot shows a web application window titled "UMS - Fee Heads Configuration". It has a purple header bar with the text "FEE HEADS". Below the header, there are three tabs: "HOSTEL FEE HEADS", "DAY SCHOLAR FEE HEADS", and "COMMON FEE HEADS". The "DAY SCHOLAR FEE HEADS" tab is selected. The main content area is titled "Day Scholar Expenditure Heads" and contains four input fields with labels and values:

Label	Value
Transport Charges:	4000
Library Fee:	2000
Sports Fee:	1500
Activity Fee:	1000

At the bottom of the form, there is a green button labeled "SAVE DAY SCHOLAR".

Figure 7: Day Scholar Fee Heads Configuration

7.4 Fee Voucher System

The screenshot shows a web application window titled "UMS - Fee Voucher Management". It has a purple header bar with the text "FEE VOUCHER SYSTEM". Below the header, there is a section titled "Generate New Voucher" with a form containing a "Student:" dropdown menu (showing "Ali Khan (2025-CS-01)"), a "Search:" input field (containing "dayscholar"), a "SEARCH" button, and a green "GENERATE VOUCHER" button. Below this section is a "Voucher List" table with the following data:

	Voucher No	Student Name	Category	Amount (PKR)	Due Date	Status
▶	V-1001	Ali Khan (2...	Hostel Stud...	66500	6/22/2026	Unpaid

At the bottom right of the form, there are two buttons: "RECORD" and "PRINT".

Figure 8: Fee Voucher Generation Form

The screenshot displays the 'FEE VOUCHER SYSTEM' interface. It includes a 'Generate New Voucher' section with fields for 'Student' (Ali Khan (2025-CS-01)) and 'Search' (dayscholar). Below this is a 'Voucher List' table with columns 'Voucher No', 'Student Name', and 'Category'. A 'Record Payment' dialog box is open, showing fields for 'Voucher No' (V-1001), 'Amount' (66500), 'Payment Date' (Sunday, June 7, 2026), and 'Payment Method' (Cash). The dialog has 'CONFIRM PAYMENT' and 'CANCEL' buttons. At the bottom of the main interface are 'RECORD' and 'PRINT' buttons.

Voucher No	Student Name	Category
V-1001	Ali Khan (2025-CS-01)	Hostel

Figure 9: Payment Recording Dialog

7.5 Reports Module

The screenshot shows the 'REPORTS CENTER' interface with the instruction 'Select a report type to generate PDF'. It features six colored buttons for different report types: 'MERIT LIST (Program-wise Merit)' in blue, 'PENDING FEE (Unpaid Vouchers)' in red, 'REFUND REPORT (Security Deposits)' in green, 'OCCUPANCY (Hostel Occupancy)' in orange, 'REVENUE SUMMARY (Category-wise)' in purple, and 'INDIVIDUAL VOUCHER (Student-wise)' in teal.

Figure 10: Reports Center

8 Sample PDF Reports

8.1 Merit List Report

UNIVERSITY MANAGEMENT SYSTEM			
MERIT LIST REPORT			
Program: BSCS			
Generated on: 07-06-2026 22:43:16			
Rank	Student Name	Merit Score	Status
1	Bilal Hassan	92.5	Eligible
2	Fatima Ali	88.3	Eligible
3	Ali Khan	85.2	Eligible
4	Sara Ahmed	78.6	Eligible
5	Usman Chaudhry	76.4	Not Eligible
Total Eligible Students: 4			
Closing Merit: 70.0			

Figure 11: Merit List PDF Output

8.2 Pending Fee Report

UNIVERSITY MANAGEMENT SYSTEM					
PENDING FEE REPORT					
Generated on: 07-06-2026 22:43:50					
Sr#	Student Name	Roll No	Amount (PKR)	Due Date	Days Late
1	Ali Khan	2025-CS-01	85,000	10-May-2026	25 days
2	Sara Ahmed	2025-CS-02	85,000	10-May-2026	25 days
3	Usman Chaudhry	2025-CS-05	85,000	10-May-2026	25 days
Total Pending Amount: PKR 255,000					
Total Defaulters: 3 students					

Figure 12: Pending Fee Report PDF Output

8.3 Individual Fee Voucher

UNIVERSITY MANAGEMENT SYSTEM

INDIVIDUAL FEE VOUCHER

Student Information

Student Name: Ali Khan

Roll No: 2025-CS-01

Program: BSCS

Category: Hostel Student

Semester: Spring 2026

Fee Breakdown

Tuition Fee:	PKR 45,000
Room Rent:	PKR 8,000
Mess Charges:	PKR 5,000
Utilities:	PKR 2,000
Maintenance:	PKR 1,500
Examination Fee:	PKR 5,000
TOTAL:	PKR 66,500

Due Date: 30-June-2026

Generated on: 07-06-2026 22:44:32

Figure 13: Individual Fee Voucher PDF Output

8.4 Revenue Summary Report

UNIVERSITY MANAGEMENT SYSTEM		
REVENUE SUMMARY REPORT		
Generated on: 07-06-2026 22:45:05		
Category	Amount (PKR)	Percentage
Tuition Fees	2,250,000	65.2%
Hostel Fees	450,000	13.0%
Transport Fees	120,000	3.5%
Examination Fees	250,000	7.2%
Other Fees	380,000	11.1%
TOTAL REVENUE: PKR 3,450,000		
Hostel Students Revenue: PKR 1,200,000		
Day Scholar Revenue: PKR 2,250,000		

Figure 14: Revenue Summary PDF Output

9 Technology Stack

Technology	Purpose
C# Windows Forms	Frontend Application
SQL Server	Database Backend
ADO.NET	Database Connectivity
iTextSharp	PDF Generation
Visual Studio	IDE
GitLab	Version Control

Table 2: Technology Stack Used

10 Login Credentials

Role	Username	Password
Admin	admin	admin123
Accounts	accounts	acc123
Student	student	student123

Table 3: System Login Credentials

Figure 15: Login Form

11 Sample Code Implementation

11.1 Login Authentication Code

```

1 private void btnLogin_Click(object sender, EventArgs e)
2 {
3     string selectedRole = cmbRole.SelectedItem.ToString();
4     string password = txtPassword.Text.Trim();

```

```
5
6     bool isValid = false;
7
8     if (selectedRole == "Admin" && password == "admin123")
9         isValid = true;
10    else if (selectedRole == "Accounts" && password == "acc123")
11        isValid = true;
12    else if (selectedRole == "Student" && password == "student123")
13        isValid = true;
14
15    if (isValid)
16    {
17        this.Hide();
18        if (selectedRole == "Admin")
19        {
20            frmAdminDashboard adminDash = new frmAdminDashboard();
21            adminDash.Show();
22        }
23        // ... similar for other roles
24    }
25 }
```

Listing 1: Login Button Click Event

11.2 PDF Generation Code

```
1 private void btnMeritList_Click(object sender, EventArgs e)
2 {
3     string fileName = $"Merit_List_{DateTime.Now:yyyyMMdd_HH:mm:ss}.pdf";
4     string filePath = Path.Combine(GetDownloadsPath(), fileName);
5
6     using (FileStream fs = new FileStream(filePath, FileMode.Create))
7     {
8         Document doc = new Document(PageSize.A4);
9         PdfWriter.GetInstance(doc, fs);
10        doc.Open();
11
12        Paragraph title = new Paragraph("UNIVERSITY MANAGEMENT SYSTEM",
13            FontFactory.GetFont(FontFactory.HELVETICA, 18, Font.BOLD));
14        title.Alignment = Element.ALIGN_CENTER;
15        doc.Add(title);
16
17        // ... add table and data
18        doc.Close();
19    }
20 }
```

Listing 2: Merit List PDF Generation

12 Expected Outcomes

The expected outcomes of this project are:

- A fully functional Windows Forms application with SQL Server backend

- Accurate total expenditure calculation for hostel and day scholar students within seconds
- Professional PDF fee vouchers with category-specific heads
- Real-time revenue breakdowns between hostel and day scholar students
- Instant hostel occupancy and expenditure reports
- Complete elimination of manual calculation errors
- Significant time and resource savings in university financial management

13 Conclusion

This University Management System focuses on accurate expenditure calculation for two distinct student categories: **Hostel Students** and **Day Scholar Students**. It automates the complex process of applying different fee heads to different student types.

- Hostel students pay rent, mess, utilities, and maintenance
- Day scholars pay transport and library fees

The system ensures that no student is overcharged or undercharged. All expenditure data is stored in separate tables for clarity and auditing. PDF reports provide transparency to both students and administration. By using C# and SQL Server, the system remains fast, reliable, and easy to maintain.

14 Future Enhancements

Possible future enhancements include:

- Integration with online payment gateways
- Email and SMS notification system
- Mobile application for students
- Advanced analytics dashboard with charts
- Automated late fee calculation
- Multi-campus support

References

1. Microsoft Documentation - Windows Forms Development
2. iTextSharp Documentation - PDF Generation
3. SQL Server Documentation - Database Design
4. C# Programming Guide - Microsoft Learn